

HW9

$$1. (a) I = \int_{-1}^3 (t^3+2)\delta(t)dt + 8 \int_{-1}^3 (t^3+2)\delta(t-1)dt$$

$$= 2 + 8(1+2)$$

$$= 26$$

$$(b) I = \int_{-2}^2 t^2 \delta(t)dt + \int_{-2}^2 t^2 \delta(t+1.5)dt + \int_{-2}^2 t^2 \delta(t-3)dt$$

$$= 0 + 2.25 + 0$$

$$= 2.25$$

$$2. (a) \text{ according to } \mathcal{L}\{\int_0^t f(x)dx\} = \frac{F(s)}{s}$$

$$\text{when } f(x) = e^{-ax} \quad F(s) = \frac{1}{s+a}$$

$$\therefore \mathcal{L}\{\int_0^t e^{-ax}dx\} = \frac{1}{s(s+a)}$$

$$(b) \int_0^t e^{-ax}dx = -\frac{1}{a}e^{-ax}\Big|_0^t = \frac{1}{a}(1-e^{-at})$$

$$\mathcal{L}\{\frac{1}{a}(1-e^{-at})\} = \frac{1}{a}(\frac{1}{s} - \frac{1}{s+a}) = \frac{1}{s(s+a)}$$

$$3. (a) f(t) = \frac{d}{dt}(e^{-at}\sin wt) \quad \mathcal{L}\{f'(t)\} = sF(s) - f(0)$$

$$\text{let } g(t) = e^{-at}\sin wt \quad G(s) = \frac{w}{(s+a)^2+w^2}$$

$$F(s) = \frac{sw}{(s+a)^2+w^2}$$

$$(b) f(t) = \int_0^t e^{-ax}\cos wx dx \quad \mathcal{L}\{e^{-ax}\cos wx\} = \frac{s+a}{s^2+(s+a)^2+w^2}$$

$$4. (a) Idc = \frac{1}{L} \int_0^t V_0 dx + \frac{V_0}{R} + C \frac{dV_0}{dt}$$

$$(b) CsV(s) + \frac{V(s)}{R} + \frac{V(s)}{Ls} = \frac{Idc}{s}$$

$$V_0(s) = \frac{Idc/c}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$$

$$(c) I_L(s) = \frac{V(s)}{Ls} \quad I_L(s) = \frac{1}{Ls} \frac{s Idc/c}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$$

$$I = sC V_0 = \frac{s Idc}{s^2 + \frac{s}{RC} + \frac{1}{LC}}$$

$$5. (a) F(s) = \frac{8s^2+37s+32}{(s+1)(s+2)(s+4)} = \frac{1}{s+1} + \frac{2}{s+2} + \frac{5}{s+4}$$

$$f(t) = (e^{-t} + 2e^{-2t} + 5e^{-4t})u(t)$$

$$(b) F(s) = \frac{20s^2+16s+12}{s(s+1)(s^2+10s+24)} = \frac{6}{s} + \frac{4}{s+2} + \frac{2}{s+4} + \frac{1}{s+6}$$

$$f(t) = (6 + 4e^{-2t} + 2e^{-4t} + e^{-6t})u(t)$$

$$(c) F(s) = \frac{20s^2+16s+12}{(s+1)(s^2+2s+5)} = \frac{4}{s+1} + \frac{8+j6}{s+1-j} + \frac{8-j6}{s+1+j}$$

$$f(t) = (4e^{-t} + 16e^{-t}\cos 2t + 2e^{-t}\sin 2t)u(t)$$

$$(d) F(s) = \frac{250(s+7)(s+14)}{s(s^2+14s+50)} = \frac{490}{s} + \frac{125\angle -163.74^\circ}{s+7-j} + \frac{125\angle 163.74^\circ}{s+7+j}$$

$$f(t) = (490 + 250e^{-7t}\cos(t-163.74^\circ))u(t)$$

$$6. (a) Z = R + sL + \frac{1}{sC} = \frac{LCs^2 + RCs + 1}{Cs} = \frac{L(s^2 + \frac{R}{L}s + \frac{1}{LC})}{s}$$

$$= 10^4 + 5s + \frac{1}{2 \times 10^{-8}}$$

$$(b) \text{ poles: } s = -\frac{R}{L} = -2000$$

$$s = \pm j\sqrt{\frac{1}{LC}} = \pm 3162j$$

$$7. I_s = \frac{15}{s} \quad L = 1H \quad I_L(s) = I(s) \frac{Z_c}{Z_L + Z_c} = \frac{15}{s} \frac{\frac{Cs}{Ls + \frac{1}{C}}}{\frac{15}{s} + \frac{Cs}{Ls + \frac{1}{C}}}$$

$$= \frac{15}{s(LCs^2 + 1)} = \frac{15}{s(s^2 + \frac{1}{LC})}$$

$$I_L(s) = \frac{150}{s(s^2+10)} = \frac{15}{s} - \frac{15s}{s^2+10}$$

$$i_L(t) = 15(1 - \cos\sqrt{10}t)u(t)$$

$$8. (a) H(s) = \frac{Ls}{R+Ls} = \frac{s}{s+\frac{R}{L}}$$

$$H(s) = \frac{s}{15000+s}$$

$$(b) \omega_c = \frac{R}{L} = 15000$$

$$(c) s = j\omega_c \quad H(j\omega_c) = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$$

$$9. f_c = \frac{1}{2\pi RC} \quad R = \frac{1}{2\pi f_c C} = \frac{1}{6.91 \times 10^{-7}} = 1447178 \Omega$$